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Roll No

EX-702-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Electrical Drives

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Draw and explain the block diagram of electric drives.
b) Draw and explain the constant torque and constant power characteristics of a.c. drives.
2. a) Explain in brief different types of braking used in electric drives?
b) Explain how regenerative braking is done in three phase induction motor.
3. a) Describe static Kramer drive for speed control of induction motor.
b) Describe static Scherbius drive for speed control of induction motor.
4. a) Describe how the speed of a d.c. series motor can be controlled by means of d.c. chopper.

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- b) A d.c. series motor, fed from 400V d.c. source through a chopper, has the following parameter
 $r_a = 0.05\Omega$, $r_s = 0.07\Omega$, $k = 5 \times 10^{-3} \text{ Nm / amp}^2$
the average armature current of 200 A is ripple free. For a chopper duty cycle of 50% determine
- Input power from the source
 - Motor speed and
 - Motor torque.
5. a) Explain the functioning of Static Scherbius Drive in detail with necessary diagram.
b) Describe stator voltage control technique using AC voltage controller for the speed control of a 3- Φ induction motor.
6. a) Explain static rotor resistance control in induction motors. Compare this scheme with slip-power recovery scheme specially at lower speed.
b) Explain the operation of a VSI fed induction motor drive.
7. a) Discuss the significance of electric braking and plugging.
b) Explain the operation of Load commutated CSI fed Synchronous Motor.
8. Write short notes on any two of the following:
- Closed loop operation of synchronous motor drives.
 - Regenerative braking for dc motor
 - Semiconverter fed dc series motor
 - Slip power recovery static Kramer drive

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